BROODBOOKS

Poultry Production Cycle Accounting System

Business Requirements Document (PRD)

Consolidated Analysis of All 11 Excel Worksheets

Ready for Manager Review & Technical Design

Version 1.0 | November 2026

Document Status: Business Analysis Completed

1. Document Control

Item	Value
Document Title	BroodBooks — Business Requirements Document (PRD)
Version	1.0
Date	November 2026
Scope	11 Excel worksheets covering full Production Cycle Accounting
Primary Users	Accountant, Manager, Veterinarian
Currency	IRR (Iranian Rial)
Date Calendar	Jalali (Solar Hijri)
Status	Business Analysis Completed — Ready for Manager Review

1.1 Purpose of this Document

This PRD consolidates the complete business analysis of the original 11-sheet Excel workbook used for poultry production-cycle accounting. It defines confirmed fields, calculation rules, business rules, functional requirements, integrations, and pending technical items for each module. The goal is to provide a single, authoritative source for implementation and manager sign-off.

1.2 Document Structure

- Section 2 — System Overview & Common Patterns
- Sections 3–13 — Detailed requirements for Sheets 1 through 11
- Section 14 — Cross-Module Integrations & Expense Categories
- Section 15 — Pending Items & Open Questions
- Section 16 — Final Status Summary

2. System Overview & Common Patterns

2.1 Business Context

BroodBooks replaces the existing multi-sheet Excel workbook that tracks an entire poultry production cycle: from chick purchases through feed, medicine, labor, mortality, sales, and final profit/loss calculations including partner shares and veterinarian bonuses based on FCR.

2.2 Common Patterns Across All Sheets

2.2.1 Automatic Sequential Row Numbering

- Field: line_no (Automatic)
- Starts at 1 and increases sequentially for each new record within a cycle.

2.2.2 Jalali (Solar Hijri) Dates

- All transaction dates use the Jalali calendar.
- Field name: event_date

2.2.3 Numeric Behavior & Display

- Underlying decimal values are preserved for all calculations.
- Display formatting follows original Excel behavior (often whole numbers).
- No unnecessary forced two-decimal restrictions unless required by business.
- Empty numeric fields treated as zero where appropriate.

2.2.4 Record Lifecycle

- Add / Edit / Delete supported on all transactional sheets.
- Summary indicators recalculate automatically after any change.
- Bulk import from Excel/CSV is MVP for most transactional sheets.
- PDF (and where specified image) attachments supported.

2.2.5 Production Cycle Association

- Every transactional record belongs to exactly one production cycle (cycle_id).

2.2.6 Seven Production Expense Categories (Sheet 10)

- 1. Chick Purchase (Sheet 1)
- 2. Medicine & Vaccines (Sheet 2)
- 3. Purchased Feed (Sheet 3)
- 4. Contract-Manufactured Feed / Toll (Sheet 4)
- 5. Raw Materials (Sheet 5)
- 6. Labor (Sheet 6)
- 7. Other Expenses (Sheet 7)

3. Sheet 1 — Chick Purchases

Status: COMPLETED and Confirmed

Module: Production Cycle Accounting | Primary User: Accountant | Entity: chick_purchase

3.1 Business Purpose

Records chick purchases for a production cycle. Captures supplier, chick type, quantity, insurance, unit price, transportation and driver. Calculates total purchase amount, landed cost and average cost indicators.

3.2 Fields (11)

#	Field	DB Name	Type	Description
1	Row Number	line_no	Automatic	Sequential number starting at 1
2	Date	event_date	Input	Jalali purchase date
3	Supplier	supplier_id	Reference	From supplier list
4	Chick Type	chick_type	Input	Type of purchased chicks
5	Quantity	quantity	Input	Whole number only
6	Insurance per Chick	insurance_per_unit	Input	Insurance cost per chick
7	Unit Price	unit_price	Input	Purchase price per chick
8	Total Purchase Amount	line_total	Calculated	Includes insurance
9	Transportation Cost	freight	Input	Transportation cost
10	Driver	driver_id	Reference	From driver list
11	Total Landed Cost	landed_cost	Calculated	Purchase + transportation

3.3 Calculation Rules

```
line_total = quantity × (unit_price + insurance_per_unit)
landed_cost = line_total + freight
Total Purchased Quantity = SUM(quantity)
Total Landed Purchase Cost = SUM(landed_cost)
Average Unit Purchase Price = Total Purchase Amount / Total Purchased Quantity
Average Landed Cost per Chick = Total Landed Purchase Cost / Total Purchased Quantity
```

3.4 Summary Fields

- Total Purchased Quantity
- Total Purchase Amount
- Total Landed Purchase Cost
- Average Unit Purchase Price
- Average Landed Cost per Chick

3.5 Key Business Rules

- Quantity must be whole number (no decimals).
- Insurance is included in line_total; transportation is excluded from line_total but included in landed_cost.
- Suppliers and drivers selected from predefined lists.

3.6 Functional Requirements (MVP)

• CHK-001 to CHK-011: Register records, auto row numbers, Jalali dates, supplier/driver selection, auto calculations, summary indicators, preserve Excel behavior.

4. Sheet 2 — Medicine & Vaccines

Status: COMPLETED — Business Analysis

Persian: دارو و واکسن | Entity: medicine_vaccine

4.1 Business Purpose

Records medicine and vaccine purchases. Stores date, supplier, category (Medicine / Vaccine / Other), quantity, unit price and transportation. Calculates purchase amounts and landed costs.

4.2 Fields (9)

#	Field	DB Name	Type	Description
1	Row Number	line_no	Automatic	Sequential
2	Date	event_date	Input	Jalali
3	Supplier	supplier_id	Reference	Predefined list
4	Medicine Type	item_kind	Input	Medicine / Vaccine / Other
5	Quantity	quantity	Input	Integer only
6	Unit Price	unit_price	Input	Price per unit
7	Total Purchase Amount	line_total	Calculated	qty × unit_price
8	Transportation Cost	freight	Input	Freight
9	Total Landed Cost	landed_cost	Calculated	line_total + freight

4.3 Calculation Rules

```
line_total = quantity × unit_price
landed_cost = line_total + freight
Summaries: SUM(quantity), SUM(freight), SUM(line_total), SUM(landed_cost)
```

4.4 Key Notes

- No separate medicine/vaccine name field.
- Quantity unit is not fixed in original Excel — no restriction introduced.
- Veterinarian assignment is handled in Sheet 11, not here.
- Supports Excel/CSV import and invoice/prescription attachments (MVP).

5. Sheet 3 — Purchased Feed

Status: Business Analysis Completed — Technical Details Pending

Persian: خوراک خریداری‌شده | Entity: feed_purchased | Related: feed_toll (Sheet 4)

5.1 Business Purpose

Records feed purchased directly from suppliers. Captures feed type, loading type, quantity (kg), unit price, transportation and driver. Costs remain separate from contract-manufactured feed, but quantities are combined for FCR and feed-per-chick calculations.

5.2 Fields (11)

#	Field	DB Name	Type	Description
1	Row Number	line_no	Automatic	Sequential
2	Transaction Date	event_date	Input	Jalali
3	Supplier	supplier_id	Reference	Predefined list
4	Feed Type	feed_type	Reference	Predefined list
5	Loading Type	loading_type	Reference	Predefined list
6	Quantity (kg)	quantity_kg	Input	Decimals allowed
7	Unit Price	unit_price	Input	Price per kg (IRR)

8	Total Purchase Amount	line_total	Calculated	qty × unit_price
9	Transportation Cost	freight	Input	Freight
10	Driver	driver	Reference	Predefined list
11	Total Landed Cost	landed_cost	Calculated	line_total + freight

5.3 Key Calculation & Integration Rules

```
Total Feed Quantity = Purchased Feed Quantity + Contract-Manufactured Feed Quantity
```

Combined quantity is used for FCR and feed quantity per chick. Costs remain separate expense categories.

- Six summary indicators including averages.
- Bulk import + PDF attachments for weighbridge/invoices (MVP).

6. Sheet 4 — Contract-Manufactured Feed

Status: Business Analysis Completed — Technical Details Pending

Persian: خوراک کارمزدی / کمیسیونی | Entity: feed_toll

6.1 Business Purpose

Records feed produced under contract-manufacturing. The unit price is the manufacturing fee (کارمزد) per kg, not the feed purchase price. Costs stay separate from purchased feed; quantities are combined for FCR.

6.2 Fields (11) — Same structure as Sheet 3 with adjusted terminology

- Manufacturing Fee per Unit (unit_price) — fee per kg
- Total Manufacturing Fee (line_total)
- Total Landed Manufacturing Cost (landed_cost)

6.3 Key Rules

```
complete_contract_feed_cost = SUM(feed_toll.landed_cost) + SUM(raw_material.landed_cost)
```

However, manufacturing costs and raw-material costs remain separate expense categories in financial reports. Raw-material quantities are NOT added to total feed quantity for FCR.

- Separate expense categories: cost_feed_purchased, cost_feed_toll, cost_raw_material

7. Sheet 5 — Raw Materials

Status: Business Analysis Completed — Some Technical Details Pending

Persian: مواد اولیه مصرفی | Entity: raw_material

7.1 Business Purpose

Records raw materials consumed in contract-manufactured feed production. Separate expense category from both purchased and toll feed.

7.2 Fields (10)

- Production Party (supplier_id — provisional mapping)
- Raw Material (material_name — predefined list)
- Item Description (item_description — free text)
- Quantity (kg), Unit Price, line_total, freight, landed_cost

7.3 Confirmed Average Landed Cost Formula

```
Average Landed Cost per Unit = Total Landed Consumption Cost / Total Quantity Consumed
(J3 / D2)
```

8. Sheet 6 — Labor

Status: Business Analysis Completed — Technical Details Pending

Persian: دستمزد | Entity: labor | Related: personnel

8.1 Business Purpose

Records personnel wages and additional payments for a production cycle. Includes days worked, daily wage, other payments (insurance etc.), and calculates total labor cost.

8.2 Fields (10)

#	Field	DB Name	Notes
1	Row Number	line_no	Automatic
2	Date	event_date	Jalali
3	Personnel	personnel_id	Reference to personnel
4	Job Description	job_task	Select existing or enter new
5	Additional Description	description	Optional text
6	Days Worked	person_days	Decimals allowed
7	Daily Wage	daily_wage	Wage per day
8	Labor Wages	wage_total	person_days × daily_wage
9	Other Payments	other_insurance	Insurance + other (combined)
10	Total Labor Cost	labor_total	wage_total + other_insurance

8.3 Summary Indicators (8)

- Total Person-Days Worked
- Other Payments to Personnel
- Number of Currently Active Personnel (automatic via is_active)
- Average Wage per Person-Day
- Average Total Cost per Person-Day
- Total Daily Wages Paid
- Total Labor Cost Paid
- Average Daily Wage Paid During Current Period (Active Personnel × Average Total Cost per Person-Day)

8.4 Personnel Status Rules

- is_active Boolean on personnel entity.
- Inactive personnel cannot be selected for new records; historical records remain.
- Manager controls Active/Inactive status.
- is_vet flag exists for veterinarian identification (bonus handled in Sheet 11).

9. Sheet 7 — Other Expenses

Status: Business Analysis Completed — Technical Details Pending

Persian: سایر هزینه‌ها | Entity: other_expense

9.1 Business Purpose

Catch-all for additional production-cycle expenses. Supports quantity × unit price + additional costs.

9.2 Fields (10)

- description, main_description, detailed_description (optional)
- quantity, unit_price → line_total = quantity × unit_price
- other_expenses (may be empty/zero) → landed_cost = line_total + other_expenses

9.3 Summary

```
Total Expenses Paid = SUM(line_total)
```

```
Total Landed Cost of Expenses Paid = SUM(landed_cost)
```

Production-cycle calculation (Sheet 10) uses the Total Landed Cost (cell J3), not the pre-additional total.

10. Sheet 8 — Mortality & Waste

Status: Business Analysis Completed — Technical Details Pending

Persian: تلفات و ضایعات | Entity: mortality

10.1 Business Purpose

Records number and weight of birds lost or discarded. Provides net chick count, mortality % and livability % for the cycle. Records are cycle-level (not flock-level).

10.2 Fields (6)

- loss_type (dropdown), chick_type (dropdown), count (integer), weight_kg (numeric)

10.3 Key Formulas

```
Net Chicks = Placed Chicks − Total Mortality & Waste Count
Mortality Percentage = Total Mortality Count / Placed Chicks × 100
Livability Percentage = 100% − Mortality Percentage
Average Weight = Total Weight / Total Count  (0 if count = 0)
```

10.4 Integration

- Placed Chicks comes from Sheet 1 SUM(quantity).
- Net Chicks used in veterinarian bonus and cost-per-chick calculations.

11. Sheet 9 — Sales

Status: Business Analysis Completed — Ready for Manager Review

Persian: جزئیات فروش دوره | Entity: sale

11.1 Business Purpose

Records sales transactions: customer, destination, quantity (kg), crates, unit price, loading cost, transportation cost, driver. Calculates gross and net sales amounts.

11.2 Fields (12)

- customer_id, destination (filtered by customer), quantity_kg, crates, unit_price
- line_total = quantity_kg × unit_price
- loading_cost, freight, driver (auto-shows vehicle)
- net_sale = line_total − loading_cost − freight

11.3 Confirmed Rules

- Selecting customer filters destinations.
- Selecting driver displays associated vehicle.
- Total number of baskets is recorded but NOT summed (not used in any formula).
- flock_id exists in project document; mandatory status not confirmed.
- Multiple attachments per sale supported.

12. Sheet 10 — Period Calculations

Status: COMPLETED

12.1 Business Purpose

Automatic financial summary of the production cycle: total revenue, selling expenses, net sales, seven production cost categories, total costs, and profit/loss (before partner & vet deductions handled in Sheet 11).

12.2 Revenue Side

```
Selling Expenses = Total Paid Loading Cost + Total Paid Transportation Cost  (from Sheet 9)
Net Sales = Total Sales − Selling Expenses
```

12.3 Production Cost Side (Seven Categories)

#	Cost Category	Source
1	Chick Purchase Cost	Sheet 1 — Total Landed Purchase Cost
2	Medicine & Vaccine Cost	Sheet 2 — Total Landed Purchase Cost
3	Purchased Feed Cost	Sheet 3 — Total Landed Purchase Cost
4	Contract Feed Production Cost	Sheet 4 — Total Landed Manufacturing Cost
5	Raw Material Consumption Cost	Sheet 5 — Total Landed Consumption Cost
6	Labor Cost	Sheet 6 — Total Labor Cost Paid
7	Other Expenses	Sheet 7 — Total Landed Cost of Expenses

```
Total Costs = Sum of all seven categories
Profit / Loss (before deductions) = Net Sales − Total Costs
```

13. Sheet 11 — Supplementary Period Calculations

Status: Business Analysis Revised (v1.2) — Ready for Final Consolidated Review

Entity: supplementary_period_calculation

13.1 Operational Indicators

Indicator	Formula / Source
Placed Chicks	Sheet 1 SUM(quantity)
Mortality & Waste Count	Sheet 8 SUM(count)
Net Chicks	Placed Chicks − Mortality Count
Mortality %	Mortality Count / Placed Chicks × 100
Livability %	Net Chicks / Placed Chicks × 100
Total Feed Consumed	Sheet 3 qty + Sheet 4 qty
Feed per Net Chick	Total Feed / Net Chicks
Total Meat Sold	Sheet 9 SUM(quantity_kg)
Average Bird Weight	Total Meat Sold / Placed Chicks
FCR	ROUNDUP(Total Feed / Total Meat Sold, 2)

13.2 Financial & Per-Unit Indicators

- Net Profit (Loss) before deductions (D6) = Net Sales − Total Costs
- Cost-Based Profit Margin = D6 / Total Costs
- Cost per Net Chick = Total Costs / Net Chicks
- Cost per Kg Sold = Total Costs / Total Meat Sold
- Live Bird Price = Net Sales per kg (after loading & transport) from Sheet 9
- Net Revenue Generated = Live Bird Price × Total Meat Sold

13.3 Partnership Settlement

- Each partner has an independent participation % for the cycle.
- Share by Weight = Total Meat Sold × Partner %
- Share by Sales = Total Meat Sold × Partner % × Live Bird Price
- Partnership Cost to Us = Total Meat Sold × Partner % × Cost per Kg Sold
- Sum of partner percentages may be ≤ 100%. Above 100% is invalid. Do NOT normalize or auto-assign remainder.

13.4 Veterinarian Bonus (FCR-based)

Bonus per Net Chick is looked up from the threshold table (H4:I20). Blank row between 1.70 and 1.71 is omitted.

FCR Threshold	Bonus per Net Chick (IRR)
1.60	20,000
1.61	16,000
1.62	15,500
1.63	15,000
1.64	14,500
1.65	14,000
1.66	13,000
1.67	12,500
1.68	12,000
1.69	11,500
1.70	8,500
1.71	7,000
1.72	6,000
1.73	5,000
1.74	4,000
1.75	0

- FCR < 1.60 → use 1.60 (20,000)
- FCR > 1.75 → use 1.75 (0)
- Within range: greatest threshold ≤ achieved FCR

```
Total Veterinarian Bonus = Bonus per Net Chick × Net Chicks
```

```
Net Profit (Loss) after deductions (D18) = D6 − Total Vet Bonus − Total Partner Share by Sales
```

13.5 Zero-Denominator Handling

If a required denominator is zero, display “Not Calculable” with a short explanation. Do not silently substitute zero.

14. Cross-Module Integrations & Expense Categories

14.1 Expense Categories in Profit & Loss

The seven categories listed in Section 2.2.6 remain strictly separate. No merging of purchased-feed, toll-feed or raw-material costs.

14.2 Feed Quantity for FCR

```
total_feed_kg = SUM(feed_purchased.quantity_kg) + SUM(feed_toll.quantity_kg)
```

Raw-material quantity is never added to this total.

14.3 Net Chicks Flow

Sheet 1 (Placed) → Sheet 8 (Mortality) → Net Chicks → used by Sheet 11 (cost per chick, feed per chick, vet bonus).

15. Pending Items & Open Questions

15.1 Reference Data Still Required

- Actual Feed Type list options (Sheets 3 & 4)
- Actual Loading Type list options (Sheets 3 & 4)
- Mortality & Waste Type dropdown values (Sheet 8)
- Chick Type dropdown values (where used)
- Production Party / Raw Material list options (Sheet 5)

15.2 Technical Design Items

- Exact database data types, numeric precision and constraints
- Driver storage mapping (string vs FK)

- Attachment storage, size limits and validation
- Bulk-import templates and validation rules
- Rounding implementation details to match Excel exactly
- Audit history for sensitive changes (e.g. FCR bonus table)
- Whether summary indicators are stored or calculated on the fly

15.3 Business Clarifications (if any)

- Whether flock_id is mandatory on Sales records
- Manager review of any remaining summary-indicator discrepancies

16. Final Status Summary

Sheet	Name	Status
1	Chick Purchases	COMPLETED & Confirmed
2	Medicine & Vaccines	COMPLETED — Business Analysis
3	Purchased Feed	Completed — Technical Pending
4	Contract-Manufactured Feed	Completed — Technical Pending
5	Raw Materials	Completed — Technical Pending
6	Labor	Completed — Technical Pending
7	Other Expenses	Completed — Technical Pending
8	Mortality & Waste	Completed — Technical Pending
9	Sales	Completed — Ready for Manager Review
10	Period Calculations	COMPLETED
11	Supplementary Period Calculations	Revised v1.2 — Ready for Final Review

All 11 sheets have completed business analysis. Remaining work consists of reference-data collection and technical design decisions. This document is ready for manager review and sign-off as the authoritative Business Requirements baseline for BroodBooks implementation.

— End of Document —